

PRACHI SINHA

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Education

Kalinga Institute of Industrial Technology Deemed to be University

2024 – 2028 (Expected)

Bachelor of Technology in Computer Science (AI Specialization)

Bhubaneswar, Odisha

- CGPA: 8.71/10

Projects

Deepfake Detection System | *Python, Scikit-learn, Pandas* **2026**

- Designed a pipeline to classify synthetic vs. authentic media using metadata features (e.g., resolution, difficulty, age range, gender).
- Implemented preprocessing and feature engineering for structured datasets.
- Built baseline ML models to establish detection accuracy and prepared the framework for future CNN/image-based upgrades.

Real Estate Price Analyzer | *Python, Scikit-learn, Pandas* **2026**

- Built a regression-based ML model to predict housing prices using features such as location, size, and amenities.
- Implemented data preprocessing, feature engineering, and model evaluation pipelines.
- Developed visualizations to interpret feature importance and pricing trends, aiding decision-making in real estate markets.

Handwritten Digit Recogniser using ANN | *Python, TensorFlow, Keras* **2026**

- Designed and trained an Artificial Neural Network to classify handwritten digits from the MNIST dataset.
- Achieved high accuracy through hyperparameter tuning and model optimization.
- Published a detailed project report and GitHub repository documenting methodology, results, and evaluation metrics.

Music Popularity Predictor | *Python, Scikit-learn, Flask, Render* **2026**

- Developed and deployed an end-to-end ML web application to predict song popularity from audio features.
- Trained and optimized a Random Forest Regressor ($R^2 \approx 0.38$, $MAE \approx 13.7$) on a dataset of 100K+ tracks.
- Implemented feature preprocessing pipeline using StandardScaler and model serialization with joblib.
- Developed interactive frontend using Flask and deployed live on Render.

Spotify Recommendation System | *Python, Pandas, Scikit-learn* **2026**

- Developed a content-based recommendation system using audio feature similarity.
- Implemented cosine similarity / distance-based ranking to suggest similar songs.
- Performed feature scaling and vector-based similarity comparison across tracks.

NLP Mini Projects (Poetic and Translation Chatbots) | *Python, HuggingFace, LLM APIs* **2025**

- Built a prompt-engineered poetic chatbot using a generative AI API to produce themed poetry.
- Implemented a multilingual translation chatbot using pretrained transformer models from HuggingFace.
- Designed interactive inference pipelines for text generation and translation tasks.

Technical Skills

Languages: Python, Java, C, SQL

ML/AI: ANN, Regression, Classification, Model Evaluation, Scikit-learn, Pandas, NumPy, TensorFlow, Keras

NLP: HuggingFace Transformers, LLM APIs

Data Visualization & Analysis: Matplotlib, Seaborn, EDA, Feature Engineering

Web & Deployment : Flask, Render, Git, GitHub

Certifications

Generative AI Program- Finlatics (2025)

AWS Cloud Practitioner Essentials- Amazon Web Services (2025)

AI Tools- Be10x (2025)